Berner Fachhochschule
Haute école spécialisée bernoise
Bern University of Applied Sciences

Detailed outline for Laurin

Memo, slides, questions and peer evaluation

Item	Details
Institution	Bern University of Applied Sciences (BFH)
Course	EAIF: AI for Finance, Autumn Semester 2026
Group	Group_8 (Beyeler Fabian, Lussi Laurin, Griessen Jan)
Document type	Working outline for the memo (3 pages plus appendix), the slides (8 minutes) and the peer evaluation
Key dates	Coaching Tue 24 Nov 2026; memo, notebook, results.json and peer evaluation Fri 27 Nov 2026, 23:59; slides Mon 7 Dec 2026, 23:59; presentation Tue 8 Dec 2026
Source of numbers	Executed solution notebook (Colab-verified), run of 3 November 2026

Purpose. Tells you what each page of the memo must contain, which numbers to use and where each is computed, wording you can edit, and what not to say. The analysis is finished.

How to use this outline. It tells you what each page of the memo must contain, which numbers to use (and where each one is computed in the notebook), wording you can edit, and what not to say. The four results sheets (tasks a to d, in the folder results_sheets/) hold the detailed numbers, figures and explanations; the plan is in docs/PLAN.md. You do not need to code. If a number looks odd, ask Fabian before you write it down.

1. Deadlines and hand-in checklist

Date	What	Notes
Sun 8 Nov	Memo skeleton in Word (the six headings) and slide skeleton (6 slides)	Walkthrough with Fabian, 30 minutes
Sun 15 Nov	Memo sections 1 to 4 drafted; outline of section 5	Use this outline
Sun 22 Nov	**Complete memo draft and complete deck**	Notebook is already frozen in content
Mon 23 Nov	Team read-through, list of questions for the coaching	Everyone reads everything
Tue 24 Nov	**Coaching with the lecturer**	Bring draft memo and deck
Thu 26 Nov	Final memo PDF; peer evaluation filled in by each of us; **submit in the evening**	Hard deadline Fri 27 Nov, 23:59
Mon 7 Dec	Final slides (PDF), internal target noon, deadline 23:59	Moodle or email to the lecturers
Sat 5 and Mon 7 Dec	Two timed rehearsals: 8 minutes	Questions afterwards: about 2 minutes
Tue 8 Dec	Presentation	Everyone answers questions on every slide

- **Four deliverables carry the group name "Group_8":** the memo PDF (name and members on page 1), the notebook, the slides PDF and results.json (the notebook writes the group name into it). Suggested file names: Group_8_memo.pdf, Group_8_notebook.ipynb, Group_8_slides.pdf, Group_8_results.json.
- **Moodle:** memo, notebook, results.json and the peer evaluation (one form per student) by Fri 27 Nov, 23:59. Slides by Mon 7 Dec, 23:59. Questions to the lecturers go to the Moodle forum.

- **Fabian** produces the final notebook run and results.json; **you** produce the memo and the slides; each of us fills in the peer evaluation.

2. What earns points (rubric) and where it goes in the memo

Criterion (points)	What the grader looks for	Where in the memo
Pipeline and data audit (20)	5: leakage-free formation panel. 15: the audit table, graded against the list of planted problems	Section 3 (compact table) and appendix A1, A2
Model comparison (20)	Linear vs tree ensemble, selection on cross-validation, test reported once, next to a naive baseline	Section 4(b), appendix A4
Where does ESG pay (15)	8: finding the cell where the premium exists and only that cell. 4: honest uncertainty. 3: point-in-time vs restated	Section 4(c), appendix A5
Backtest with costs and floor (15)	Formed after publication, held 12 months, costs charged, cost of the floor shown, bootstrap of the difference to the benchmark	Section 4(d), appendix A6
Memo and slides (15)	Clear question, clear answer, figures with labelled axes and units, no overclaiming; 10-minute talk (we plan 8)	Whole memo, the slides
What would change with real data (15)	5: the list of what remains after the audit. 10: which conclusions would survive and which not, and why	Section 5 (about 0.6 page)

Three rules the handout states: every metric appears next to the baseline it must beat (the naive average forecast for models, the equal-weight benchmark for the backtest); every figure has labelled axes with units; every feature is dated. **Also:** state that the data is simulated and cite it with its licence on the memo and on the slides: "Simulated ESG and stock-return dataset, EAIF course, Autumn Semester 2026, CC BY 4.0", https://github.com/umatter/EDFB/tree/main/data/esg2026.

3. Memo: page budget (3 pages including tables and figures, plus an appendix of any length)

The lecturer confirmed that tables and figures count towards the three pages. Use 10 pt text, 2 cm margins, 8 pt tables, and put everything else in the appendix with a pointer ("see Appendix A4").

Page	Content	Space
1	Title block (Group_8, members, date), 1 Question and short answer, 2 Data, start of 3 Audit: compact audit table (7 rows)	Title 4 lines; Question 5 lines; Data 8 lines; table 0.4 page
2	3 Pipeline: decision rule, split, feature table (compact), 4(b) model comparison (text + 4-row table), start of 4(c)	Pipeline 0.2 page; feature table 0.25; (b) 0.3; (c) start 0.2
3	4(c) region table (compact), 4(d) text + 4-row table, 5 What would change with real data, 6 Group members and roles	(c) 0.2; (d) 0.25; real data 0.45; roles 3 lines
Appendix	A1 full audit table; A2 full feature table; A3 figures a1, b1, b2; A4 tuning grid and full (b) table; A5 figures c1, c2 and full (c) table; A6 figure d1 and the robustness tables; A7 how to reproduce (Colab link, run order)	No limit

If page 3 is too full, shorten section 5 by moving the "list" into the table of conclusions, but keep both halves: 5 points are for the list, 10 for which conclusions survive and why.

4. Memo, section by section

Title block (page 1, top)

- "Group_8: Can ESG scores and machine learning earn an investor a premium? A memo to the investment committee". Members: Beyeler Fabian, Lussi Laurin, Griessen Jan. Date. One line: "All data are simulated.".

1. The question (5 lines, plus a two-line answer)

- **Question:** "Can an investor use ESG scores, or forecasts built on them, to earn higher returns, and can the data be trusted for that purpose?" Four sub-questions: is the data reliable, can a model forecast 12-month returns better than the average, where (if anywhere) did the Environmental score pay, and does a rule with costs and an ESG floor beat an equal-weight benchmark?
- **Short answer (committee readers love this):** "The data needed seven fixes, one of which (a leaking column) would have produced a false forecasting edge. No model beat the naive forecast. The Environmental score was associated with higher returns in one place only, Europe from 2021, on four years of data. A backtest with costs showed no edge that can be told apart from noise. We would not allocate on this evidence."

2. The data (8 lines)

- Simulated by the course, no real firms. Five files: firm-year panel (14,250 rows, fundamentals, point-in-time ESG scores, environmental metrics), restated ESG scores (as a current download would show them), monthly returns (171,764 firm-months), monthly factors, and the current universe (987 survivors). 1,376 firms, fiscal years 2012 to 2025; firms enter and leave (389 exits).
- Licence line, exactly: "Simulated ESG and stock-return dataset, EAIF course, Autumn Semester 2026, CC BY 4.0", with the link.
- Say in one sentence that simulated data cannot show how ESG behaves in real markets; it tests our method. (This returns in section 5.)

3. The audit and the pipeline (page 1 bottom to page 2 top)

Audit (compact table, three columns: problem / treatment / effect on one number). Use short phrases; the long version is appendix A1. Rows and numbers (source: notebook (a) Step 10):

Problem	Treatment	Effect on one headline number
28 firms: revenue and market cap about 1,000x too large (probably USD thousand)	Divide by 1,000	Mean market cap 213,205 to 14,108 USD million
300 near-duplicate return months	Keep first row	Firm-years with a full 12 months: 9,002 to 9,182
Analyst_Outlook carries future information (rank correlation 0.68 with the next 12 months, about 0 with other windows)	Exclude from all models	Random forest test error 37.6 without it, 29.1 with it (baseline 37.2)
Survivorship: universe file has 987 of 1,376 firms	Use the full panel, hold exits to their last month	Equal-weight benchmark 4.62% (full panel) vs 5.18% (survivors only)
Restated scores contain hindsight	Point-in-time scores everywhere; restated only for the comparison	Within-year rank correlation with next excess return 0.010 vs 0.056
Carbon, water, energy missing together (13.2% of rows), not at random	Keep rows, carbon intensity filled plus a missing flag	Linear test error 38.10 to 38.05 without carbon features
Scores clipped at 0 or 100 (213 rows, 1.5%), 8 returns at -95%	Keep, document	Linear test error 38.10 to 38.00 without clipped rows

- **Say how each was found in one phrase** (the detail is in appendix A1): consistency of carbon, water and energy with revenue; duplicate keys; correlation with the return in three 12-month windows; file sizes; comparison of the two score files; missing share by region and score; counts at the bounds. Add the sentence from the notebook: "Each effect was measured one problem at a time, so the effects do not add up."
- **Pipeline (5 to 6 lines).** Portfolios are formed on 1 July of each year; every feature comes from the fiscal-year row published between 31 March and 31 May (known before 1 July) or from returns up to June; the target is the total return over the next 12 months. Split by formation year: training 2015 to 2021, test 2022 to 2024 (the test set is used once); cross-validation by company inside the training years. Mention the decision-time check built into the

code: nothing is used before its publication date.

- **Feature table (compact, 8 pt):** group the given features in one row ("given: growth, margin, four ESG scores, size, past 12-month return, industry and region"), then one row each for the six new features with the date from which each is known: Env_Change, Margin_Change, log_MC_to_Rev, log_Carbon_Int, Carbon_Missing (all from the fiscal-year row, known by 31 May of the formation year) and Past_Vol_Pct (returns through 30 June). Analyst_Outlook as the row "excluded". Full table: appendix A2 (notebook (a) Step 9).

4(b). Model comparison (about 0.35 page)

- **Text:** use the memo paragraph on results sheet (b). Put the four-row table next to the baseline: baseline 37.20 test RMSE; linear 38.10; random forest 37.93; gradient boosting 37.84; CV 29.16 / 28.88 / 28.92 / 28.90.
- **The one sentence the handout wants:** "The extra complexity of the ensembles does not pay off out of sample."
- **Must-have honesty notes:** (1) the selection of linear regression is effectively a tie (within 0.06 points); (2) the test years were a +21% boom (against +0.9% in training), so cross-validation (28.9) looks far better than the test (37 to 38); the starter notebook's "downturn" is wrong for this data; (3) after removing year averages the models still do not beat zero, rank correlation about -0.02 and changing sign.
- **Translate for the committee:** an error of 37 to 38 points on 12-month returns means these forecasts cannot size or rank positions.

4(c). Where does ESG pay (about 0.4 page)

- **Text:** use the memo paragraph on results sheet (c). The deliverable is one table (region by period, point-in-time and restated) and one sentence. Compact table: 7 regions plus "All regions", four columns, coefficient with standard error in brackets, stars. Full table with intervals and p-values: appendix A5.
- **The sentence (copy from results.json):** "An investor could have earned an ESG premium only in European firms from 2021 (+3.1 points of 12-month excess return per standard deviation of the Environmental score, Holm-adjusted p = 0.018); no other region or period is distinguishable from zero, the estimate rests on only 4 formation years and is a cross-sectional association, not a return shown to be tradable after costs."
- **Honest uncertainty (4 points):** 1 of 14 cells; four formation years (p about 0.09 with years as the unit); intervals not adjusted for the 14 cells; the premium shows from 2022 (2021 is +0.35).
- **Point-in-time vs restated (3 points):** restated scores give higher coefficients everywhere and more significant cells, because the restatement behaves as if it contains hindsight (rank correlation 0.16 with the next return). Say "behaves as if", not "contains".

4(d). Backtest with costs and an ESG floor (about 0.3 page)

- **Text:** use the memo paragraph on results sheet (d). Compact table with the four rows (no floor, floor 50th, floor 80th, benchmark): return, volatility, Sharpe, turnover. Figure d1 goes to appendix A6; refer to it.
- **The sentence:** "The floor did not cost return in this sample (it cost concentration and trading), and the best rule's Sharpe advantage over the benchmark (+0.11, 95% interval -0.16 to +0.38) is not distinguishable from noise."
- **Must-have notes:** the model used is the one selected in (b) and has almost no ranking skill (rank correlation +0.01); the 80th-percentile floor holds about 19 names; the clean out-of-sample years 2022 to 2024 tell the same story; gradient boosting and a linear variant are labelled robustness checks; costs are the starter's 20 bp convention (40 bp on the replaced share); benchmark is equal weight, no cost, full panel including exits.

5. What would change with real data (about 0.6 page; 15 points). See the detailed content in the next section.

6. Group members and their roles (3 lines)

- Beyeler Fabian: notebook (data lead and modelling lead). Lussi Laurin: writing lead (memo and slides). Griessen Jan: **role to be agreed by the group before the memo is finished** (the handout asks for roles to be named; a natural fit is

checking tables and figures against the notebook). Close with the attribution line again.

5. Detailed content for "What would change with real data"

5a. The list: what remains to be solved after an audit like ours (5 points)

Write these as a compact list, one line each; the rightmost column is evidence from our own work you can quote.

What remains unsolved on real data	Why it matters	Evidence in our work
ESG scores are proprietary and raters disagree	Our data has one score per firm. Real providers use different methods and their scores correlate only moderately (published research reports an average correlation around 0.5 between major raters: Berg, Koelbel and Rigobon 2022; verify the figure before citing). A result for one rater may not hold for another.	Restated vs point-in-time: 86% overlap in the top decile, so 14% of the names change
Point-in-time availability	Vendors revise and backfill scores. Without dated vintages we cannot know what an investor saw on 1 July. We had a publication date and a separate restated file; real data rarely does, and an Analyst_Outlook-type column would be hard to spot.	Restated scores behave as if they contain hindsight; Analyst_Outlook: test error 37.6 to 29.1
Survivorship that is harder to see	Our universe file betrayed itself (987 of 1,376 firms). Real databases silently drop delisted firms, and delisting returns (bankruptcy, takeover) are often missing.	Survivors-only benchmark overstates the return by about 0.6 points a year; -95% returns not in the last month
Costs are not a constant	We used 20 bp each way. Real costs depend on spread, size, liquidity, market stress and the size of the order; a concentrated 19-name portfolio is harder to trade.	Turnover 0.70 to 0.82; cost drag 0.28 to 0.33 points a year
Units, identifiers and duplicates	We found a unit error and duplicates through cross-column checks. Real data has mergers, name changes, currency and restated fundamentals that are harder to detect.	28 firms off by a factor of 1,000; 300 duplicated months
Missing ESG data is not random	Companies choose what to disclose. Missingness related to the score and to region biases any sample that drops gaps, and it is lowest in Europe, the one region where we found an effect.	13.2% missing; Europe 6.3% vs 13 to 19% elsewhere
Short samples, one regime, many looks	14 years, a three-year test boom, 14 region-period cells. Real data is longer but the risk of a false discovery remains.	Test years +21.6% vs +0.9%; Europe result rests on four years (p about 0.09)

5b. Which of our conclusions would survive, which not, and why (10 points)

This is the larger part. Use a table: conclusion, survives or not, why, and what would test it on real data.

Our conclusion	On real data	Why	What would test it
No model beats the naive forecast out of sample	**Likely survives in direction**	Stock returns are weakly predictable; the signal-to-noise ratio is low everywhere. With real data and more years a small skill could appear, so the claim is "no evidence of skill", not "impossible".	Time-forward cross-validation over many years; several regimes; more features
The specific data problems (unit error, duplicates, a leaking column)	**Do not survive as such**	They were planted. Real data has other problems. What survives is the procedure: consistency checks across columns, correlation with returns in several time windows, publication dates, comparison with a survivors-only file.	Run the same checks on the real vendor files and ask the vendor for documentation
Survivors-only and restated data flatter results	**Direction survives, size does not**	Both mechanisms are real; their size depends on the exit rate, delisting returns and vendor practice. The 0.6 points a year is specific to this simulation.	Point-in-time constituent lists, delisting returns, data vintages

Our conclusion	On real data	Why	What would test it
Environmental score paid in Europe from 2021 (+3.1 points per SD)	**Does not survive as an investable conclusion**	The simulation planted one relationship; we found it. In real data a single cell out of 14, on four years, would need replication before anyone relied on it. The result may also reflect disclosure rules or regional events we cannot see here.	Out-of-sample replication on other years, other raters and other regions; a pre-registered hypothesis
Restated scores overstate the premium	**Likely survives in direction**	Backfilled or revised scores can embed information that arrived later; any test on them is optimistic. The size cannot be known without vintages.	Compare strategies built on original versus revised scores
Backtest: no edge distinguishable from the benchmark	**Survives, and net results get worse**	No edge before real costs means no edge after them. Real costs exceed our flat 20 bp, especially for a concentrated portfolio.	Backtest with spread- and size-dependent costs and capacity limits
The ESG floor did not cost return	**Does not survive**	With 19 to 47 names the difference is noise, and in our data the score is unrelated to returns outside one cell. In real markets a floor shrinks the universe, concentrates the portfolio and raises tracking error.	Compare with random floors of the same size; report tracking error
Cross-validation looks better than the test because the test years were a boom	**Survives as a lesson**	Models should be judged across regimes; random cross-validation mixes years and is optimistic.	Walk-forward evaluation over several market regimes

Closing recommendation (two lines): "We would not allocate on this evidence. Before relying on any ESG signal we would need several raters, dated vintages, delisting returns, realistic costs and an out-of-sample replication of the one finding."

6. Where every number in the memo is computed in the notebook

A memo result that the notebook does not reproduce earns no points. Notebook file: group_project_AS2026/notebook/Group_Work_AS_2026_solution.ipynb (Colab badge at the top). Cells are headed "(a) Step n" and so on.

Memo number	Notebook cell
28 firms, market cap 213,205 to 14,108, figure a1	(a) Step 2
300 duplicates, 9,002 to 9,182	(a) Step 3
Analyst_Outlook 0.68, test error 37.59 vs 29.14	(a) Step 4
Survivorship 4.62% vs 5.18%, 389 exits, 302 dropped firm-years	(a) Step 5
Restated: 459 firms, 0.010 vs 0.056, 86% overlap	(a) Step 6
Missing 13.2%, Europe 6.3%; scores at 0 or 100: 213 rows	(a) Step 7
Panel counts (10,410 rows, 30 features, 6,473 train, 2,709 test)	(a) Step 8
Feature table with dates	(a) Step 9
Audit table with effects	(a) Step 10
Model table, tuning grid, figures b1 and b2	(b) Steps 1, 2 and 3
Printed reading of (b): within-year check, rank correlation by year	(b) Step 5
Region by period table, figure c1	(c) Steps 2 and 3
Restated vs point-in-time, joint regression	(c) Step 4
Europe in detail, year by year, years-as-unit p, figure c2	(c) Step 5
One-sentence conclusion (c)	(c) Step 6
Backtest table, robustness, 2022 to 2024 only	(d) Step 2

Memo number	Notebook cell
Bootstrap intervals	(d) Step 3
Cumulative return figure d1	(d) Step 4
results.json (group name, members, all numbers)	Final cell at the end of the notebook

7. Slides: 6 slides for 8 minutes (plus about 2 minutes of questions)

The lecturer asked for one slide that explains the audit. The slide count and software are free, but 8 minutes is tight: six slides, about 1 to 1.5 minutes each. Every slide carries the dataset citation in a small footer line. Figures can be taken from results_sheets/figures/ (labelled axes, units).

Slide	Title and content	Time	Say (key sentences)
1	**Question and data.** The question; "all data simulated"; five files, 1,376 firms, 2012 to 2025; citation and licence	1:00	"Can ESG scores, or models built on them, earn an investor a premium? We tested this on a simulated panel, so the exercise tests our method, not the market."
2	**What we found in the data and what we did (the audit slide).** Compact table of 5 to 7 problems (problem, fix, effect); pipeline strip: 1 July rule, split 2015-2021 / 2022-2024, 30 dated features; figure a1 optional	1:30	"Seven problems. The most dangerous is a column that carries next year's return: a model using it cuts the test error from 37.6 to 29.1, but no investor had it on 1 July, so we excluded it."
3	**No model beats the naive forecast.** The four-row table or figure b1 (left); line: test years +21% boom, rank correlation about -0.02	1:30	"The models learn something in training but nothing that carries over. Complexity does not pay off."
4	**ESG paid in one place: Europe from 2021.** Figure c1 or the 14-cell table with Europe from 2021 highlighted; line: 1 of 14 cells, four years, restated scores would show more cells	1:15	"+3.1 points per standard deviation, adjusted p 0.018. One cell, four years: plausible, not proven. Restated scores contain hindsight and would show more."
5	**Backtest with an ESG floor.** Figure d1 (left) or the four-row table; line: Sharpe difference +0.11, interval -0.16 to +0.38	1:30	"The floor did not cost return here; it cost concentration and trading. The edge over the benchmark is not distinguishable from noise."
6	**What would change with real data.** The four named issues (raters disagree, point-in-time, survivorship, costs) and the survive / does not survive split; closing recommendation	1:15	"The method survives; the specific findings do not. We would not allocate on this evidence."

- **Rehearse twice with a timer** (Sat 5 and Mon 7 Dec). If you run over, cut slide 2 detail, never slide 6.
- Every group member must be able to answer questions on every slide: after each walkthrough, write three sentences in your own words per task.

8. Questions the lecturer may ask (and short answers)

Question	Short answer
Why divide by 1,000 and not drop the firms?	The environmental columns look normal for their size, so only revenue and market cap are wrong. Dropping loses 28 firms for nothing; the lecturer accepted any consistent, explained method.
How do you know Analyst_Outlook leaks?	Rank correlation 0.68 with the next 12 months and about 0 with the two earlier windows, and a test error that falls from 37.6 to 29.1 when it is used. We say it behaves as if it were stamped with future information; the data cannot show how it arose.
Why is the test error higher than the cross-validation error?	The test years were a boom (+21.6% against +0.9%), and cross-validation mixes years. Even after removing year averages the models show no skill.
Why linear regression?	Lowest cross-validation error, but a tie within 0.06 points. We followed the rule set in advance and report gradient boosting as a labelled check.

Question	Short answer
Is the Europe result real?	It is the only cell that survives a multiple-testing correction, but it rests on four years (p about 0.09 with years as the unit) and is an association, not a tradable return.
Why Holm?	14 cells were examined. A false hit is likely by chance, so we raise the bar; only Europe from 2021 clears it.
Why is the restated version bigger?	What the restatement added correlates with the next return (rank correlation 0.16): it behaves as if it contains hindsight.
Did the ESG floor cost anything?	Not return in this sample. It cost concentration (19 names at the 80th percentile), volatility and trading.
Is the benchmark comparison fair?	Equal weight on the full panel including firms that left, no cost, as in the starter; using survivors only would raise it by about 0.6 points a year.
How realistic are 20 bp?	A simplification. Real costs depend on spread, size and market conditions, so net results would be lower.
Could the notebook be rerun?	Yes: Colab, Runtime, Restart session and run all; a few minutes; fixed seeds and fixed cross-validation folds.
What did you use AI for?	AI tools are encouraged by the lecturer. We used them for coding and reviewing; every group member can explain every cell. Keep a short log.

9. Style and quality checklist (the lecturer's bar)

- **Check every claim before writing it.** Only write "beyond X" or "controlling for Y" if the notebook ran that comparison. If unsure, ask Fabian.
- **Translate every number into what it means for an investor** (for example: an error of 37 points means the forecast cannot size positions).
- **Be precise with magnitude words** and check them against the numbers: "slightly", "much", "unusually". Avoid "unusually strong" for the test years (2018 was as strong); say "all three test years average about +21%".
- **Say "associated with"**, not "earned" or "caused", for the Europe result; say "behaves as if" for the restated scores and Analyst_Outlook.
- **State limitations plainly:** four years, 14 cells, one regime, 84 months, small tuning grids, flat costs.
- **Every metric next to its baseline; every figure with labelled axes and units; every feature dated.**
- **Keep numbers consistent:** one decimal convention (for example returns with one decimal, Sharpe ratios with two); use the numbers on the results sheets, not rounded versions from memory.
- **Write for an investment committee:** short sentences, no code names (say "Environmental score", not Environmental_Score), explain RMSE once as "typical forecast error".

10. Peer evaluation and open items

- **Peer evaluation:** every student rates the contribution of every group member (including themselves) in the Moodle form, by the memo deadline (Fri 27 Nov, 23:59). Where ratings and the presentation show clearly unequal contributions, the group score is adjusted for the individuals concerned.
- **To agree as a group:** the role wording for the memo (section 6), who presents which slide, and the rehearsal times.
- **Waiting on the lecturers:** the exact notebook deadline (will be close to the hand-in date in November) and confirmation that the talk is 8 minutes.
- **Fabian will tell you** if any number in the notebook changes; the results sheets then need a quick check.